Lesson 5.3 **Duration: 40 minutes**

Conditional Logic

Learning Objectives

- Understand if-then and if-then-else blocks in Scratch
- Create programs that make decisions based on conditions
- Apply conditional logic to real-world problem scenarios

Engage (5-7 minutes)

Every day, farmers make decisions: if it rains, irrigate the fields; otherwise, wait. If the crop price is high, sell; otherwise, store. Conditional logic in Scratch teaches computers to make similar if-then decisions, like a farmer's wisdom applied to programming.

Explore (10-12 minutes)

Students experiment with sensing blocks that return true or false. They connect conditions to action blocks, creating programs where sprites react differently based on touches, key presses, or variable values — like a farmer adjusting activities based on weather conditions.

Explain (10-12 minutes)

Conditional logic uses Boolean expressions (true/false) to determine actions. The "if-then" block executes code only when a condition is true. The "if-then-else" block offers two paths — one for true, one for false. Like a river splitting around a boulder, the program flows in different directions based on conditions.

Elaborate (8-10 minutes)

Students create a project where a sprite changes color when touching the mouse pointer. Like a chameleon changing colors based on surroundings, their sprite responds to environmental conditions using if-then blocks.

Evaluate (5-8 minutes)

Can students explain how if-then blocks work? Can they create programs with at least two different conditions? Do they understand Boolean logic concepts?